

An exact quadratic-chaos reduction for Heilman's Question 1.1 and the sign-paired spectral subcase

Automated mathematical discovery note (GPT5.6)

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Status. Likely valid partial result. The general asymmetric-spectrum case remains open in this note.

1 Source question

Let γ_n be standard Gaussian measure on $\mathbb{R}^n$. In the Introduction of [1], Question 1.1 asks, without a constraint on $\gamma_n(A)$, whether

$$F(A) \leq \frac{1}{\pi}$$

for every measurable symmetric set $A = -A$. After rotating coordinates so that the centered second-moment matrix is diagonal, the functional in the paper is

$$F(A) = \sum_{i=1}^n \left(\int_A (1 - x_i^2) d\gamma_n(x) \right)^2.$$

Define the coordinate-free matrix

$$C_A := \int_A (I - xx^\top) d\gamma_n(x).$$

Orthogonal invariance and diagonalization give $F(A) = \|C_A\|_F^2$, where $\|\cdot\|_F$ denotes the Frobenius norm.

2 Idea of the proof

Frobenius duality turns $\|C_A\|_F$ into an integral of a centered Gaussian quadratic form. For a fixed quadratic form, its positive and negative sets are themselves symmetric, and choosing one of them captures exactly half of the form's first absolute moment. This gives an exact equivalence, not only a one-sided reduction.

For a spectrum made of opposite pairs, the characteristic function of the quadratic form is real and positive. The elementary inequality $\log(1+s) \leq s$ compares it pointwise with the characteristic function e^{-t^2} of a Gaussian with variance two. The standard Fourier formula for $\mathbb{E}|Z|$ then proves the desired constant.

3 Exact reduction

Lemma 1 (Best set for a centered even function). *Let $q \in L^1(\gamma_n)$ be even and satisfy $\int q d\gamma_n = 0$. Then*

$$\sup_{A=-A} \left| \int_A q d\gamma_n \right| = \frac{1}{2} \int_{\mathbb{R}^n} |q| d\gamma_n.$$

The supremum is attained, up to null sets, by $\{q > 0\}$ and by $\{q < 0\}$.

Proof. Write $q = q_+ - q_-$. The mean-zero assumption gives $\int q_+ d\gamma_n = \int q_- d\gamma_n = \frac{1}{2} \int |q| d\gamma_n$. For every measurable A ,

$$- \int q_- d\gamma_n \leq \int_A q d\gamma_n \leq \int q_+ d\gamma_n.$$

Since q is even, both sign sets are symmetric and attain the endpoints. $\square$

Theorem 2 (Exact quadratic-chaos formulation). *If $X \sim N(0, I_n)$, then*

$$\sup_{A=-A} \|C_A\|_F = \frac{1}{2} \sup_{\substack{U=U^\top \\ \|U\|_F=1}} \mathbb{E} \left| X^\top U X - \text{tr } U \right|.$$

Consequently, Heilman's Question 1.1 is equivalent to

$$\mathbb{E} \left| X^\top U X - \text{tr } U \right| \leq \frac{2}{\sqrt{\pi}} \quad (U = U^\top, \|U\|_F = 1). \quad (1)$$

For a fixed U , a maximizing set in the inner optimization is a quadratic threshold set.

Proof. For symmetric matrices use the Frobenius inner product. Put $q_U(x) = x^\top U x - \text{tr } U$. Then q_U is even, Gaussian-integrable, and centered, while

$$\langle C_A, U \rangle_F = \int_A \text{tr}(U(I - xx^\top)) d\gamma_n(x) = - \int_A q_U(x) d\gamma_n(x).$$

Therefore, by finite-dimensional norm duality and Lemma 1,

$$\begin{aligned} \sup_{A=-A} \|C_A\|_F &= \sup_{A=-A} \sup_{\|U\|_F=1} |\langle C_A, U \rangle_F| \\ &= \sup_{\|U\|_F=1} \sup_{A=-A} \left| \int_A q_U d\gamma_n \right| \\ &= \frac{1}{2} \sup_{\|U\|_F=1} \mathbb{E} |q_U(X)|. \end{aligned}$$

The sign sets of q_U are the asserted quadratic thresholds. Finally, $F(A) = \|C_A\|_F^2$, so the bound $F(A) \leq 1/\pi$ for every A is equivalent to (1). $\square$

4 The sign-paired spectral subcase

Theorem 3 (Opposite-pair spectra). *Let $U = U^\top$, $\|U\|_F = 1$, and suppose that the eigenvalue multiset of U is*

$$(a_1, -a_1, \dots, a_m, -a_m, 0, \dots, 0).$$

Then

$$\mathbb{E} \left| X^\top U X - \text{tr } U \right| < \frac{2}{\sqrt{\pi}}.$$

In particular, the non-strict inequality in (1) holds for this entire spectral class.

Proof. By rotational invariance, with independent standard Gaussians G_j, H_j ,

$$Q := X^\top U X - \text{tr } U \stackrel{d}{=} \sum_{j=1}^m a_j (G_j^2 - H_j^2).$$

The Frobenius normalization says $2 \sum_j a_j^2 = 1$. Independence gives the real positive characteristic function

$$\phi_Q(t) = \prod_{j=1}^m (1 + 4a_j^2 t^2)^{-1/2}.$$

Since $\log(1 + s) \leq s$ for $s \geq 0$,

$$\log \phi_Q(t) = -\frac{1}{2} \sum_j \log(1 + 4a_j^2 t^2) \geq -2t^2 \sum_j a_j^2 = -t^2.$$

Thus $\phi_Q(t) \geq e^{-t^2}$. For a centered integrable random variable with real characteristic function, the first absolute-moment identity is

$$\mathbb{E}|Q| = \frac{2}{\pi} \int_0^\infty \frac{1 - \phi_Q(t)}{t^2} dt.$$

It follows that

$$\mathbb{E}|Q| \leq \frac{2}{\pi} \int_0^\infty \frac{1 - e^{-t^2}}{t^2} dt = \frac{2}{\sqrt{\pi}},$$

where integration by parts gives the final integral as $\sqrt{\pi}$. If $U \neq 0$, then at least one a_j is nonzero and $\log(1 + s) < s$ for $s > 0$; hence the characteristic-function comparison is strict for every $t \neq 0$, which makes the final inequality strict. $\square$

Corollary 4 (A class of symmetric sets). *Suppose the nonzero eigenvalues of C_A occur in opposite pairs. Then $F(A) \leq 1/\pi$, strictly if $C_A \neq 0$. In dimension two, this applies whenever $\text{tr}(C_A) = 0$.*

Proof. If $C_A = 0$ there is nothing to prove. Otherwise set $U = C_A / \|C_A\|_F$. Its spectrum is sign-paired, and

$$\|C_A\|_F = |\langle C_A, U \rangle_F| \leq \frac{1}{2} \mathbb{E}|q_U(X)| < \frac{1}{\sqrt{\pi}}$$

by Lemma 1 and Theorem 3. Square the inequality. For a two-by-two symmetric matrix, trace zero is equivalent to its eigenvalues being $a, -a$. $\square$

5 Verification, limitations, and literature status

The proof above is finite-dimensional and analytic; no numerical statement is used. A Monte Carlo subgradient search over dimensions up to twenty found no violation of (1). It approached $2/\sqrt{\pi}$ through diffuse positive spectra, consistent with the central limit theorem, but this is only supporting evidence.

For a general spectrum,

$$\phi_Q(t) = \prod_i e^{-it\lambda_i} (1 - 2it\lambda_i)^{-1/2}$$

has a phase. Its real part need not dominate e^{-t^2} pointwise, so the argument of Theorem 3 does not extend directly. This is the precise remaining obstruction.

Bounded searches used the arXiv id, the exact wording of Question 1.1, and phrases including “second Wiener chaos L^1 ”, “mean absolute deviation Gaussian quadratic form”, and “weighted centered chi-square absolute moment”. No paper explicitly resolving Question 1.1 was found. However, Theorem 3.1 of [2] proves the same Gaussian first-moment upper bound for every symmetric infinitely divisible distribution. Since the sign-paired Q above is symmetric and infinitely divisible, that theorem independently subsumes the moment inequality in Theorem 3. It does not state the exact set/quadratic-chaos reduction or identify Heilman’s matrix-functional subcase. Novelty is therefore limited and should be assessed by a specialist.

References

- [1] S. Heilman, *Low Correlation Noise Stability of Symmetric Sets*, arXiv:1511.00382 (2015).
- [2] L. B. Klebanov, A. V. Kakosyan, and I. V. Volchenkova, *Inequalities for m -Divisible Distributions and Testing of Infinite Divisibility*, arXiv:1904.07604 (2019), Theorem 3.1.